Do you Know?



Feeler Gauge is an instrument which is used to measure the air gap between the stator and the rotor of the motors. The air gap value may be from few mils to 50 mills. 1 ml is equal to 0.025 mm or 1/1000 inch. Generally, the air gap is value of d.c. machines is more than a.c. machines.

10.3 Faults in a power system

Definition

Fault in electrical equipment or apparatus is defined as an imperfection in the electrical circuit due to which current is deflected from the intended path. In other words, the fault is the abnormal condition of the electrical system which damages the electrical equipment and disturbs the normal flow of the electric current.

The most common and dangerous fault that occurs in a power system is the short circuit or shunt fault. During short circuit fault, heavy current flow through the circuit which damages the insulation of current carrying phase conductors corresponding to earth or in the insulation between phases.

Types of electrical fault

Types of electrical fault can be classified into two major type

i. Symmetrical fault
ii. Asymmetrical fault

Symmetrical fault

The faults are of symmetrical in nature and occurs to symmetrical current, i.e., equal fault current in all the three phases with 120° displacement.

Types of fault

a. Ground fault in all three phases
b. Short circuit fault between phases

a. Ground fault in all three phases

It occurs due to a breakdown of insulation between all the phases as well as to the earth. It is the most severe type of the fault and rarely occurs in the power system.

b. Short circuit fault between phases

It mainly occurs due to a breakdown of insulation between all the three phases. It occur is rarely 2% to 3% in the power system.

Asymmetrical fault

This fault are asymmetrical nature and occurs to asymmetrical current, i.e., different currents in the three phases.

Types

i. Single Phase to ground fault
ii. Phase-to-Phase fault
iii. Two phases to ground fault
iv. Phase to phase and third phase to ground fault

1. Single phase to ground fault

i. It is also called a line-to-ground fault.
ii. It mainly occurs due to insulation breakdown between one of the phase and earth.

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